Can AI do the trick?

Emily C. Zabor
Department of Quantitative Health Sciences

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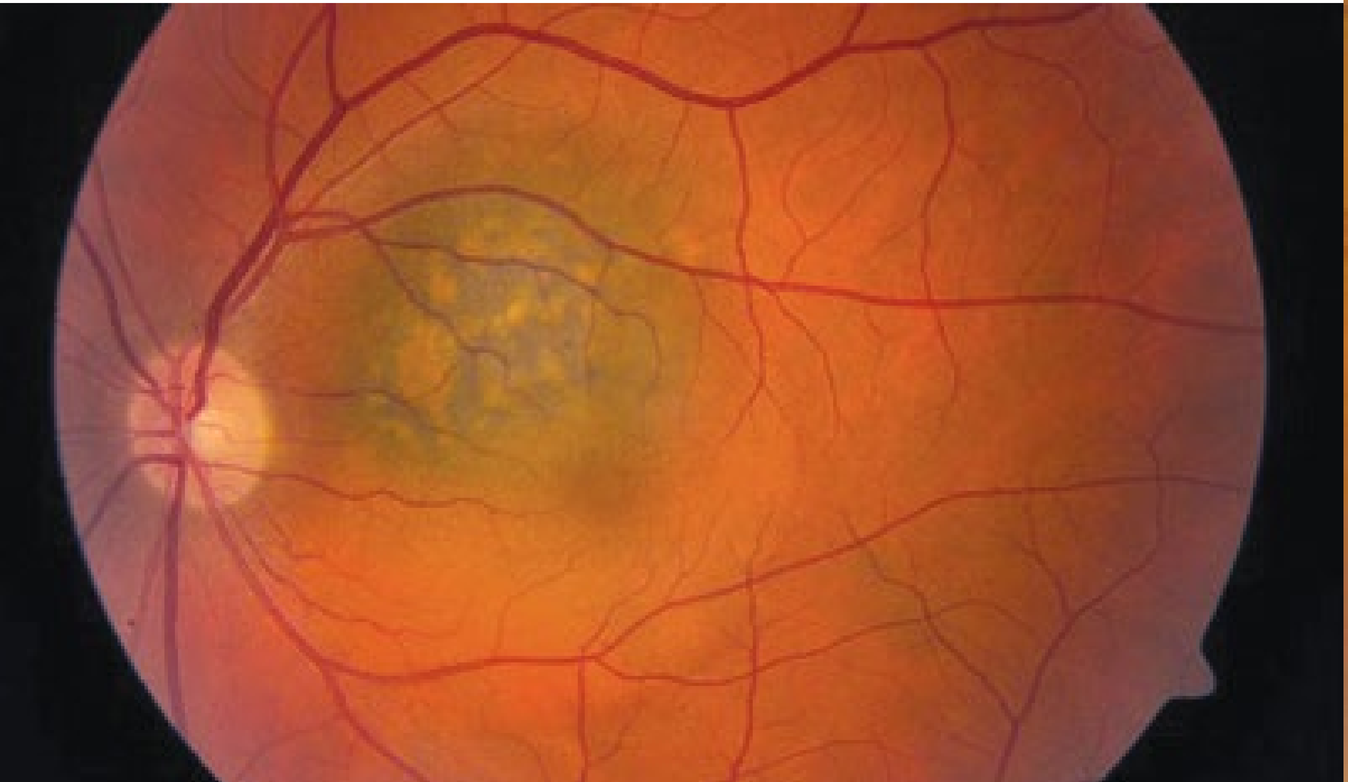
With indeterminate choroidal melanocytic tumors, how do we distinguish nevus from melanoma?

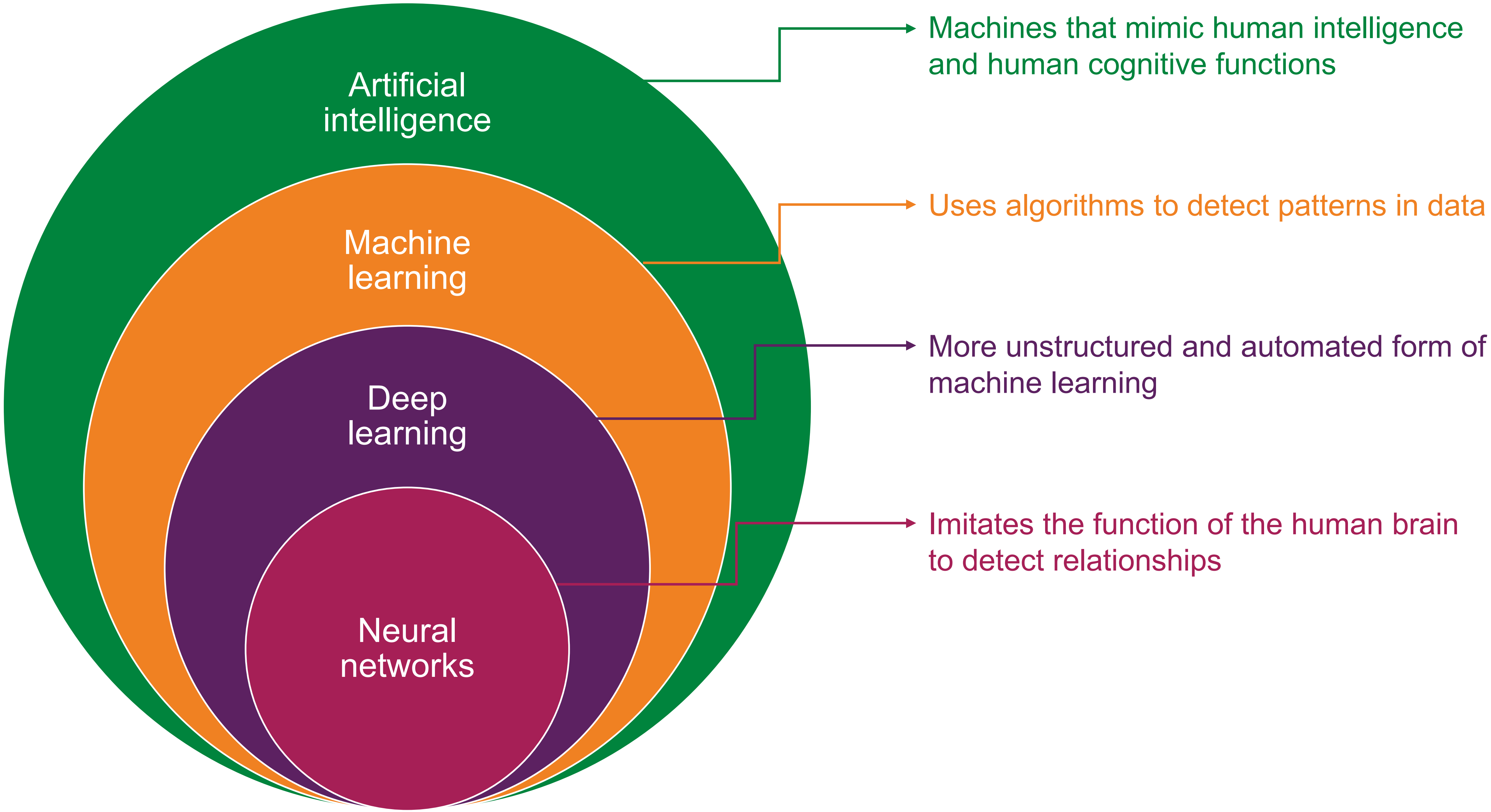
Typically rely on predictors of future growth like orange pigment and subretinal fluid.



https://www.aao.org/eyenet/article/distinguishing-choroidal-nevus-from-choroidal-mela

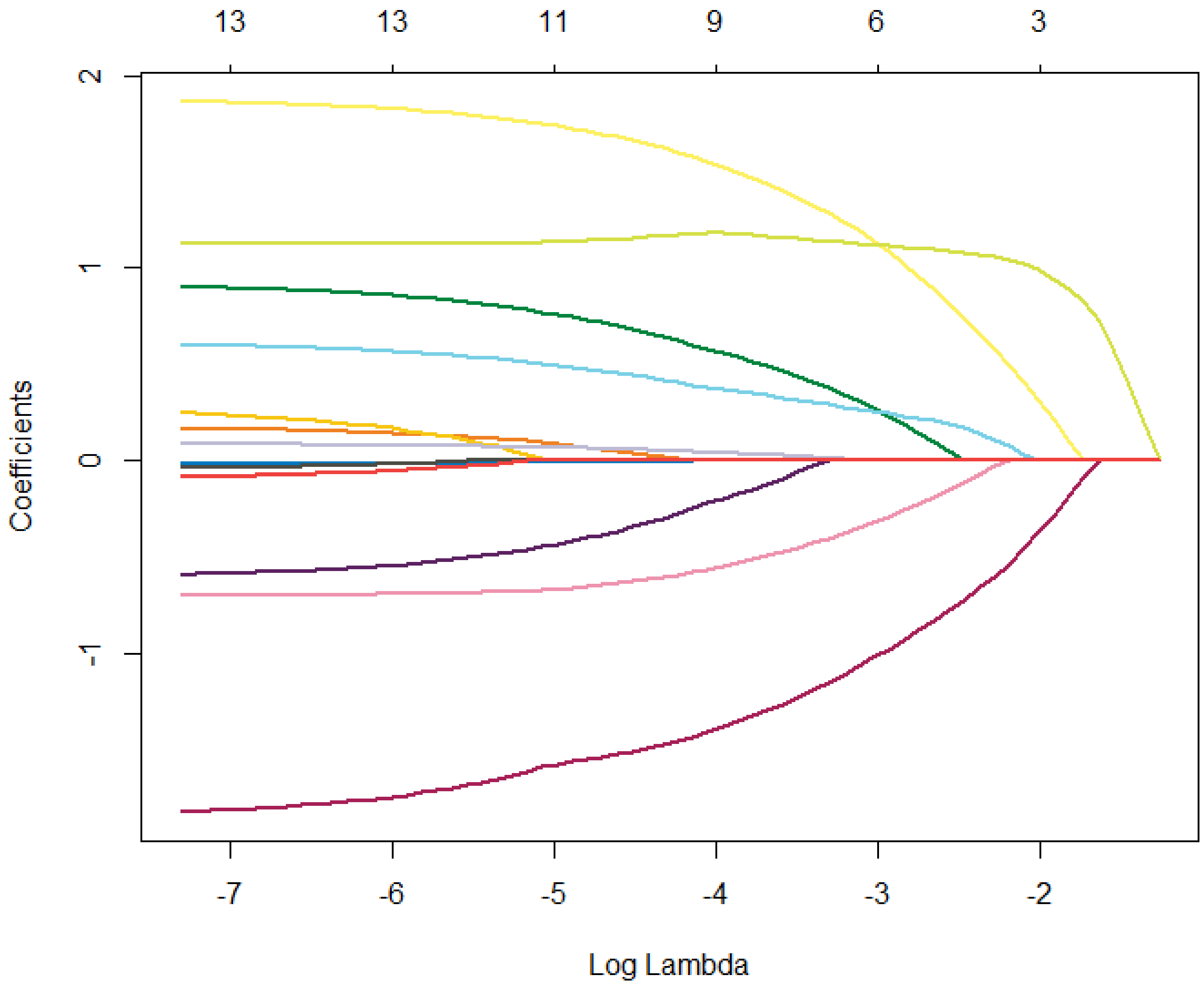
Can we do better with artificial intelligence?

https://www.esrcheck.com/2023/06/05/artificial-intelligence-ai-experts-sign-statement-on-ai-risk/

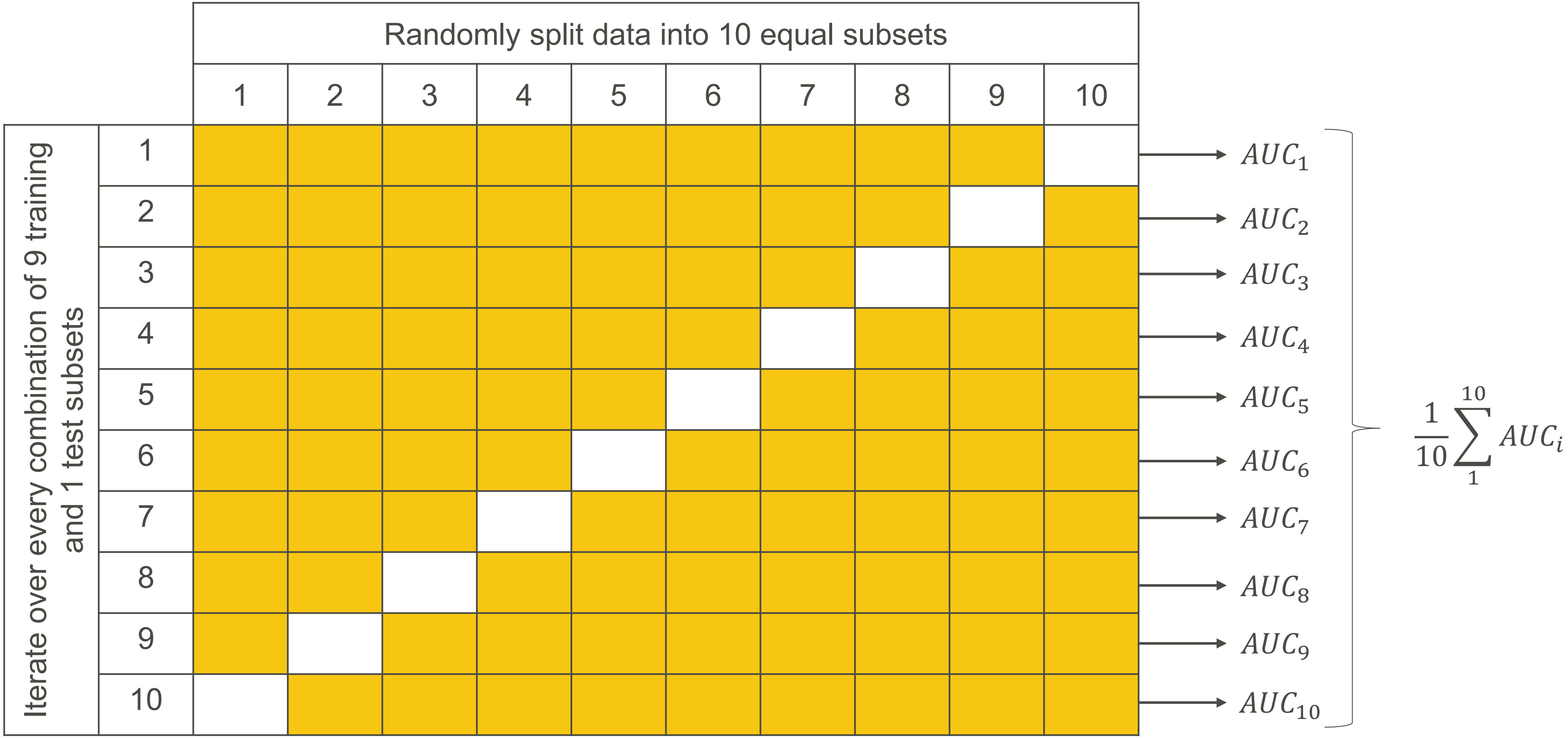


Artificial intelligence
Machine learning
Deep learning
Neural networks
Machines that mimic human intelligence and human cognitive functions
Uses algorithms to detect patterns in data
More unstructured and automated form of machine learning
Imitates the function of the human brain to detect relationships

Lasso regression is a machine learning algorithm that can be used to identify useful predictor variables



For each lambda value, calculate the cross-validation AUC. Then select the lambda that maximizes AUC.



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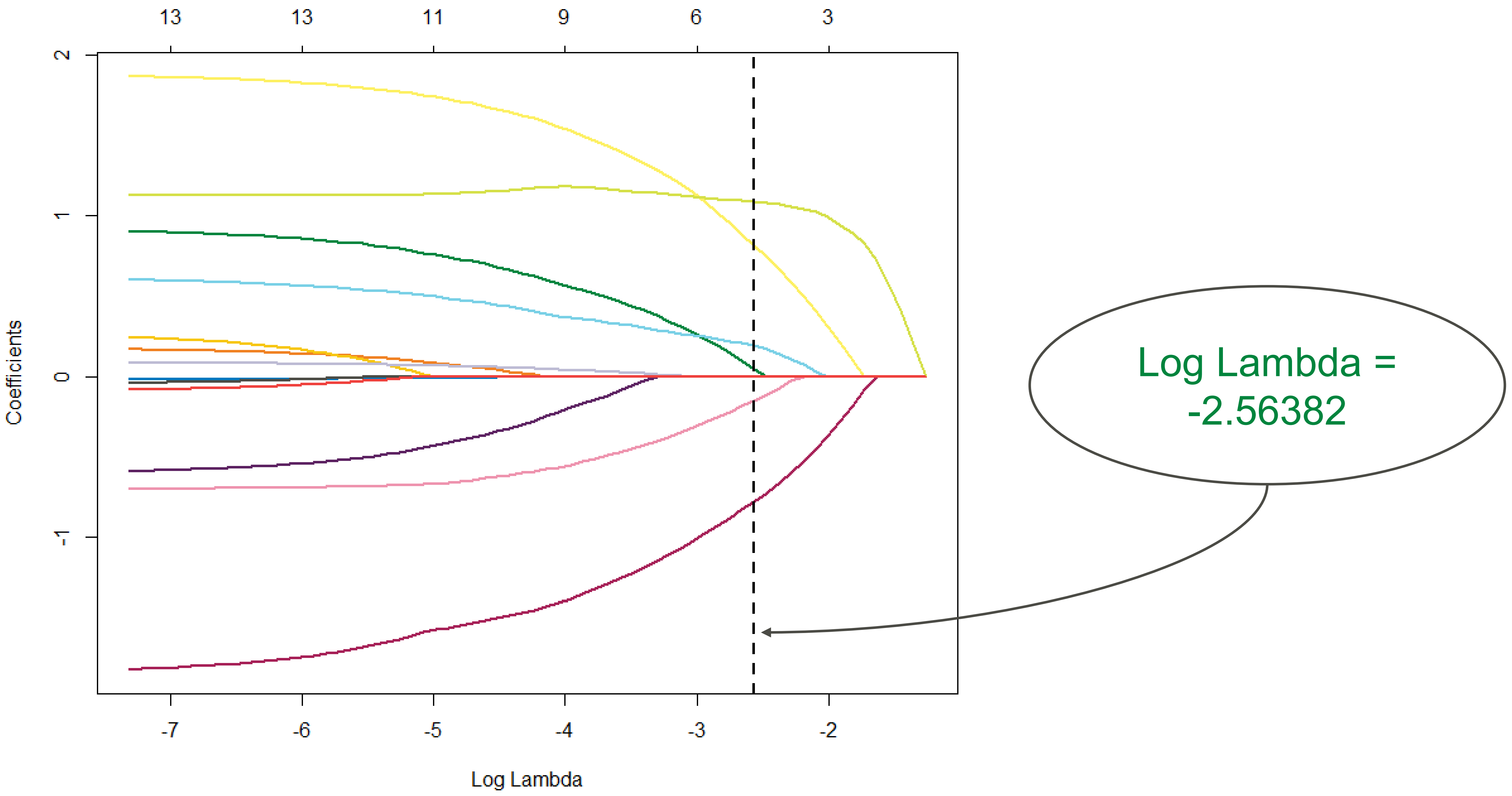
A Prediction Model to Discriminate Small Choroidal Melanoma from Choroidal Nevus

Emily C. Zabor[a] Vishal Raval[b] Shiming Luo[b] David E. Pelayes[c] Arun D. Singh[b]

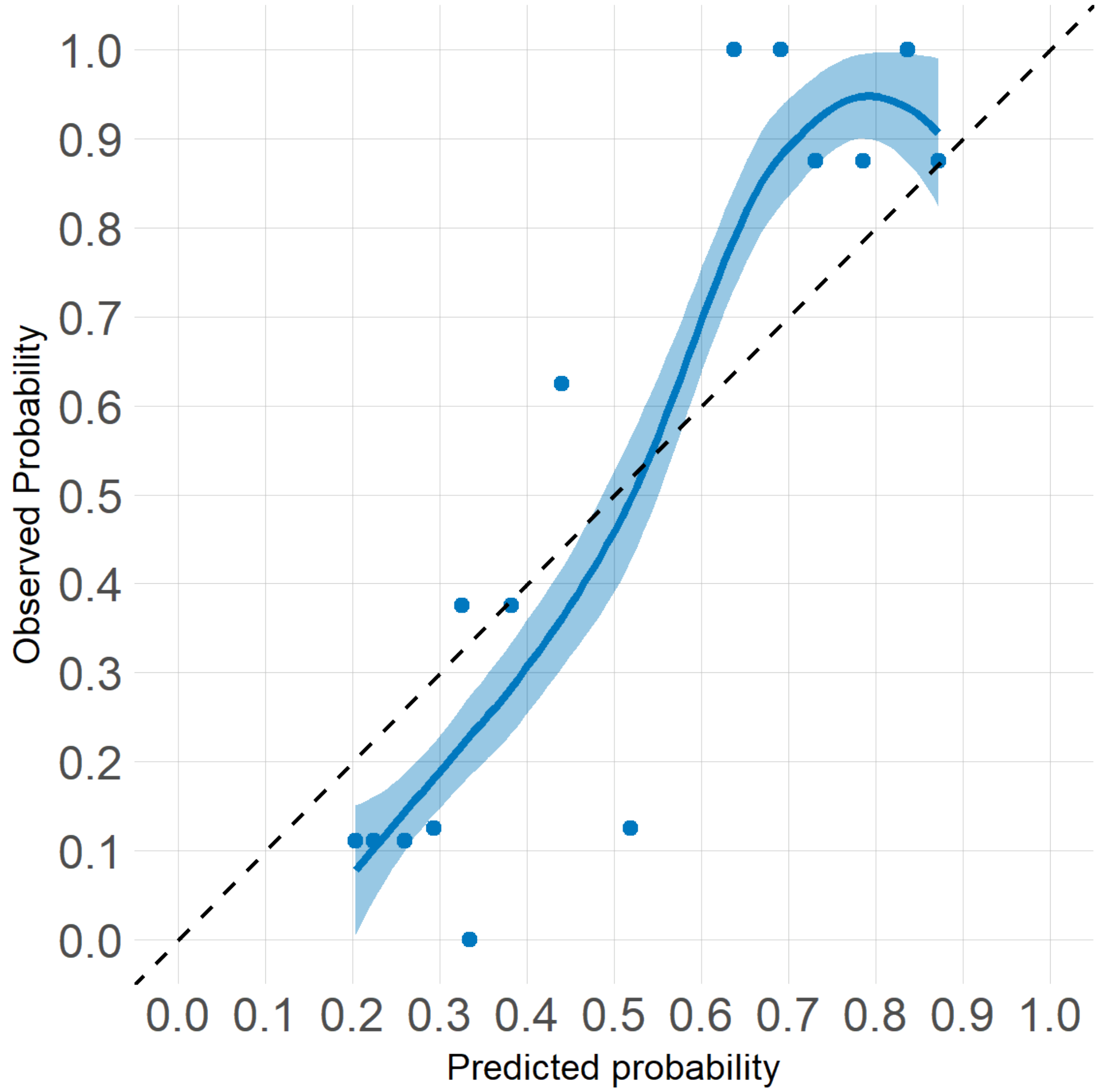
[a]Taussig Cancer Institute, Department of Quantitative Health Sciences, Cleveland Clinic, Cleveland, OH, USA; [b]Cole Eye Institute, Ophthalmic Oncology, Cleveland Clinic, Cleveland, OH, USA; [c]Department of Ophthalmology, Buenos Aires University and Maimonides University, Buenos Aires, Argentina

Characteristic	Nonmelanoma ($N = 62$)[1]	Melanoma ($N = 61$)[1]	p value[2]
Age at diagnosis	64 (56, 71)	61 (55, 67)	0.2
Gender			
Female	44 (71)	25 (41)	0.002
Male	18 (29)	36 (59)	
Laterality			
Left eye	30 (48)	31 (51)	>0.9
Right eye	32 (52)	30 (49)	
Symptoms	9 (15)	24 (39)	0.004
Distance from optic disc			
<3 mm	6 (9.7)	33 (54)	<0.001
≥3 mm	56 (90)	28 (46)	
Distance from fovea			
<3 mm	16 (26)	30 (49)	0.013
≥3 mm	46 (74)	31 (51)	
Largest basal diameter, mm	7.50 (6.50, 9.00)	9.00 (7.50, 10.50)	0.003
Height, mm	1.50 (1.30, 1.80)	2.10 (1.70, 2.30)	<0.001
SRF	9 (15)	43 (70)	<0.001
Orange pigment	19 (31)	47 (77)	<0.001
Drusen	51 (82)	26 (43)	<0.001
RPE atrophy	27 (44)	17 (28)	0.10

SRF, subretinal fluid; RPE, retinal pigment epithelium. [1] Statistics presented: median (IQR); *n* (%). [2] Statistical tests performed: Wilcoxon rank-sum test; χ^2 test of independence.

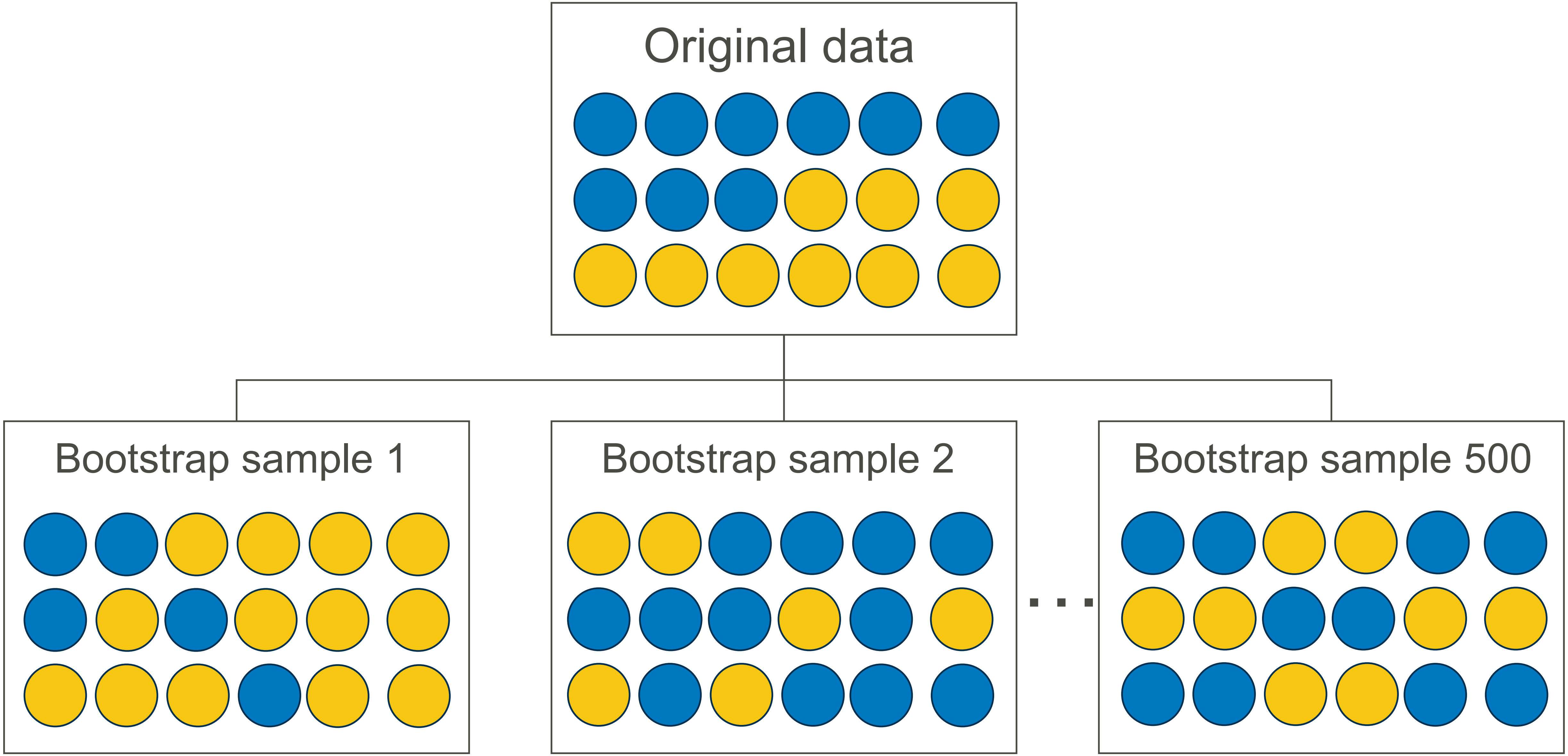

13
13
11
9
6
3
2
1
0
-1
Coefficients
-7
-6
-5
-4
-3
-2
Log Lambda
Log Lambda =
-2.56382

Characteristic	Estimate	Odds ratio
Intercept	−1.413	0.24
Male	0.047	1.05
Distance to optic disc ≥3 mm versus <3 mm	−0.776	0.46
Height, mm	0.814	2.26
Subretinal fluid	1.089	2.97
Orange pigment	0.190	1.21
Drusen	−0.148	0.86



Apparent AUC = 0.880

Internal validation using bootstrap samples with replacement

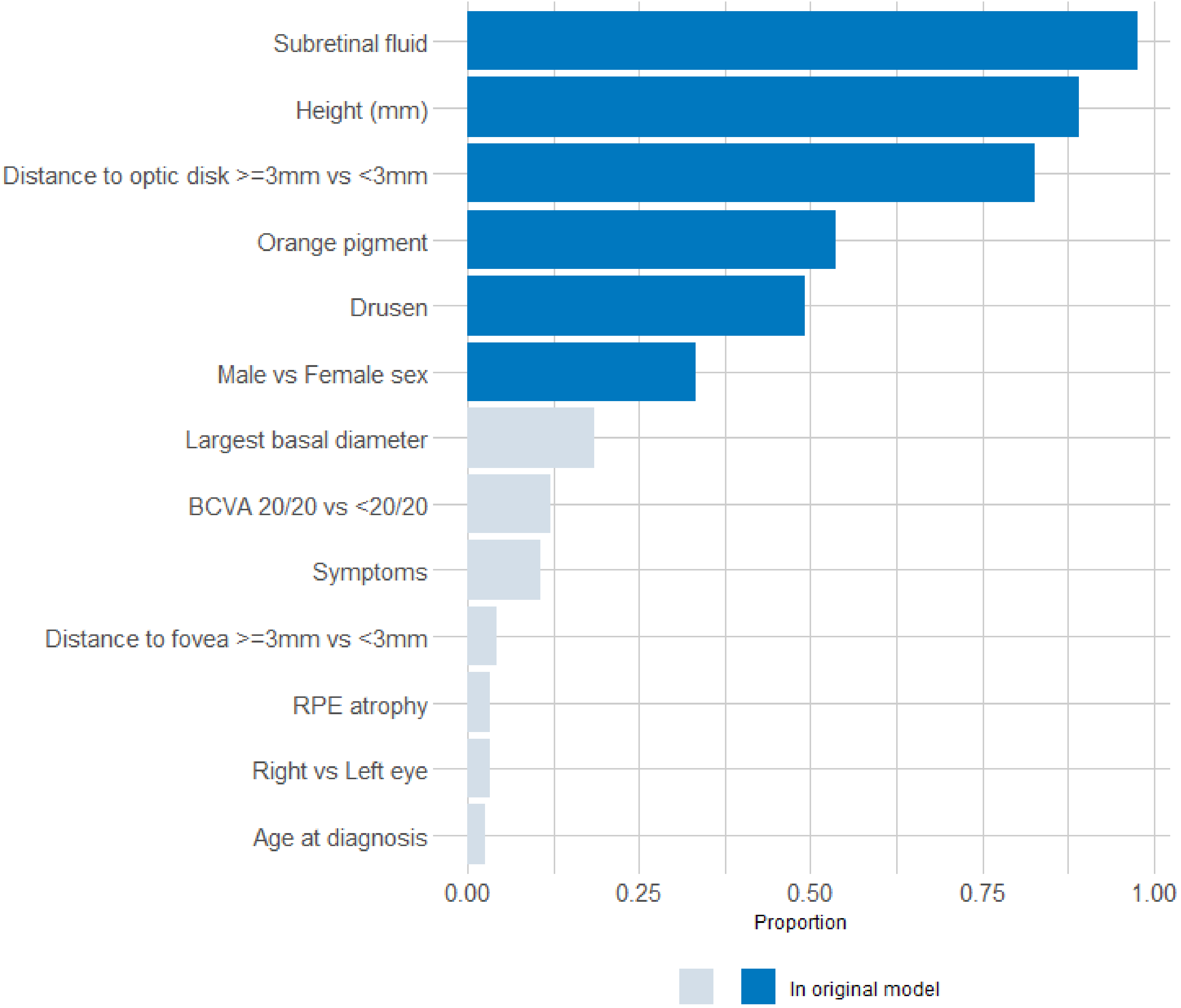


On each bootstrap dataset:

1. Repeat cross-validation procedure to select lambda
2. Refit the lasso model
3. Calculate AUC on bootstrap sample
4. Apply model to original data
5. Calculate AUC on original data

Optimism = bootstrap AUC – original AUC

Optimism-corrected AUC = 0.849



http://www.riskcalc.org/SCMprediction/

Small Choroidal Melanoma Predictor

This predictor is applicable to patients with small choroidal melanocytic lesions (i.e. at least 1mm and not more than 3.5mm in height and/or at least 1mm and not more than 15mm in largest basal diameter).

Distance to optic disk (mm)

- (•) <3mm
- ( ) >=3mm

Sex

- (•) Male
- ( ) Female

Height (mm)

1 — 1.5 — 2.5

1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.5

- [ ] Subretinal fluid
- [ ] Orange pigment
- [ ] Drusen

The predicted probability of melanoma at present is 46.4%.

Reference

[1] Zabor, E. C., Raval, V., Luo, S., Pelayes, D. E., & Singh, A. D. (2022). A Prediction Model to Discriminate Small Choroidal Melanoma from Choroidal Nevus. Ocular oncology and pathology, 8(1), 71–78. https://doi.org/10.1159/000521541

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Thank you! Questions?

zabore2@ccf.org